

Psychological Evaluation System: A Neuro-Inspired Psychometric Assessment Prototype for Workplace Crisis Resilience

Akshithaa H

Dept. of Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India
230701025@rajalakshmi.edu.in

Farheen Tabassum H

Dept. of Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India
230701088@rajalakshmi.edu.in

Dr. K. Ananthajothi

Professor, Dept. of Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India
anathajothi.k@rajalakshmi.edu.in

Abstract—Behavioral interviews and stress tests in the workplace rely on subjective evaluation of results; two panels might give different interpretations of the same answer, and a candidate that is prepared for the test will be more resilient than he really is. In this paper, we discuss the implementation of the Psychological Evaluation System – a prototype that intends to provide a better consistency of decision-making processes. The candidate provides a response to the believable scenario of a crisis in the workplace using the web-based interface. The response is analyzed using a semantic analyzer that is based on Gemma model implemented on Google Colab or on local rule-based analysis in case when Colab is not available. The result is six psychometric parameters based on Perceived Stress Scale, the State-Trait Anxiety Inventory and the Saliency Style Inventory. The six variables are then converted into seven values corresponding to each of Yeo 7 networks and are further calculated into one resilience index by a multi-layer perceptron (MLP). The seven network values are interpretive estimates normalized for the target data; our system neither scans nor measures anyone's brain, and all of its output is labeled non-clinical. With six held-out samples in k-fold cross-validation, the MLP produced $R^2 = 0.976$ with respect to its calibration targets and the whole process took median 1.18 s in live operation or 0.02 s in local fallback mode. Criterion validity, fairness testing and neuroimaging validation will be carried out in future work.

Keywords—psychometric assessment, Yeo 7-network model, behavioral resilience, explainable AI, stress tolerance, multi-layer perceptron, large language model, human-in-the-loop evaluation

I. INTRODUCTION

A problem common to recruiters and managers alike is that the interviewee who gives an excellent interview does not necessarily perform well in a difficult situation. The reflective interviewee might come across as rather quiet during conversation while the eloquent interviewee might be frozen when he encounters a problem for the very first time. Behavioral interviews are designed to predict behavior during pressure situations, but their results depend upon the tiredness of the interviewer and the ability to persuade of the interviewee [1].

Stress resilience is especially difficult to determine by the conversation. The validated psychological tests such as the Perceived Stress Scale (PSS) [3] and the State-Trait Anxiety Inventory (STAI) [4] require a person to define his or her self in abstract terms but give little indication of how a person thinks

through a particular stress situation. The written reply to the real situation gives more information because the way a person organizes the reply, who he or she includes in the situation, how actions are ordered and uncertainty is addressed, reveals much.

This work makes such an answer a matter that can be considered by a committee. This answer has been decomposed into six meaningful psychometric indices and mapped into the seven major brain networks based on the Yeo parcellation [5] and then passed through an MLP and summarized in a resilience score with language interpretation. The application of network jargon is based on network neuroscience, which considers executive and attention networks for regulation of the response and limbic, somatomotor, and default-mode networks for arousal and rumination [6], [7].

What the system is doing is made clear. The system does not scan, record, nor measure any brain; the seven network values are software-derived scores that provide biological motivation to the score, and not neuroimaging outcomes. Everything the system produces is labelled "non-clinical demo result" on-screen. The contributions are on the level of system and workflow: (i) a transparent chain of transformations from text to indicator to network to score where each intermediary value is displayed; (ii) a two-mode architecture where, if the hosted Gemma model is available, it is used, and if not, the deterministic local analyser is employed to ensure that the result is produced; and (iii) a candidate-oriented UI with no clues and an evaluator-oriented console providing complete audit trail. The rest of the paper is organised as follows: Section II covers the literature review, Section III introduces the system, Section IV the implementation and results, and Section V the conclusions.

II. LITERATURE SURVEY

Review of the structured employment interview by Levashina et al. [1] demonstrated that standardization of questions and rating scale enhance inter-interviewer consistency; while decades of selection research as well as limited validity of selection method was summarized by Schmidt and Hunter [2]. This suggests the use of a constant situation with rating scale for explicit evaluation. Cohen et al. [3] created the PSS scale measuring unpredictability, uncontrollability and overload perceived by a person in his/her

life; while Spielberger [4] divided state and trait anxiety in the STAI test. Six variables used here were adapted from these measures and applied to the text form.

The work by Yeo et al. [5], in which seven networks were identified based on resting-state connectivity from a sample of one thousand subjects, is used as the standard reference parcellation now. The approach to describe psychopathology as interactions of several major networks was suggested by Menon [6]; Arnsten [7] explained the effect of stress signaling on prefrontal regulation, while Raichle et al. [8] discussed the default mode of brain operation associated with self-referential thinking. Such facts allow grouping the psychological symptoms based on network names, but not predicting brain activity from an essay of an individual.

Sentence embedding models that represent short texts as numeric vectors to use in other models include Sentence-BERT [9]. Models that generate brain activity predictions for video, audio, and language stimuli from the population of people who were scanned in the process include TRIBE v2 [10], which predicts cortical responses based on population-level data; however, the current research is inspired by the idea of representing brain activity using the Yeo networks of such a model. Adedokun et al. [11] describe human-in-the-loop systems and trust calibration in the context of agency-based AI, while Cheng and Cheng [12] offer decoupled architecture for human intervention control, both providing justification for an architecture that involves informing humans with the model output. The comparison of these approaches with the proposed one is presented in Table I.

TABLE I
COMPARISON WITH RELATED APPROACHES

Approach	Free text	Psycho-metric	Network view	Explain-able	HITL
Structured interviews [1], [2]	—	—	—	—	✓
PSS / STAI [3], [4]	—	✓	—	—	—
Brain-encoding models [10]	✓	—	✓	—	—
HITL frameworks [11], [12]	—	—	—	—	✓
This work	✓	✓	✓*	✓	✓

* Interpretive network estimates, not measured neural activity.

Taken altogether, these papers have studied interview validity, stress measurement, brain structure, and human supervision individually. The gap in all of these is the lack of an inspectable pipeline that transforms a person's natural language reaction during a crisis situation into the corresponding measures based on the existing constructs and expressed in the language of networks, compares the result

against reference situations, and shows everything to the human panel honestly.

III. PROPOSED SYSTEM AND METHODOLOGY

A. Problem Statement and Research Questions

The challenge is to build a decision support system that analyzes the candidate's textual input to a crisis situation at the workplace and outputs an interpretable resilience profile to the interview committee. Four research questions have been defined to guide the research process. RQ1: Is it possible to project a textual answer to a small set of meaningful psychometric features? RQ2: Is it possible to build a regression model which maps psychometric features into Yeo-network estimates and reproduces its calibration targets? RQ3: Is it possible to maintain dual mode operation to retain the availability and validity of the output in the case of inaccessible hosted model?

B. System Architecture

There are three modules within the prototype, as depicted in Fig. 1: a browser-based interface (index.html) which displays the scenario and captures the answer, providing a live console; a Flask backend (server.py listening on port 5000) which implements GET /api/scenarios and POST /api/submit endpoints and controls the pipeline in llm.py; and an analysis service (colab_server.py) running on Google Colab which provides Gemma analysis and MLP prediction via a Localtunnel URL. When the user submits the response, the backend tries to contact the Colab service first, allowing up to ten seconds. If this request succeeds, its results are utilized; otherwise, the same steps are followed using local deterministic computations. In both cases, the outcome is checked against the benchmark file, translated into interpretation and sent back to the interface while being printed in a special banner from the server command line. The backend relies on Python 3.11, Flask with CORS extension, NumPy, scikit-learn (MLPRegressor module) and the requests library; tribev2.txt (scenario and seven baseline attributes) and tribev2_output.csv (benchmark values) serve as the knowledge base.

Figure 3.1: Tribe V2 Neuro-Cognitive System Architecture

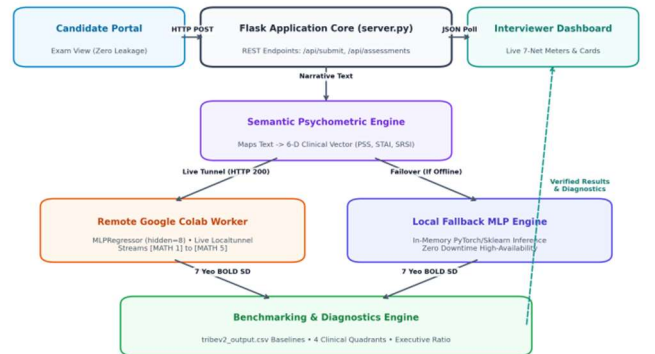


Fig. 1. System architecture of the Psychological Evaluation System: candidate portal, Flask application core, live Colab and local fallback engines, and the benchmarking and diagnostics engine.

C. Semantic Analysis and Psychometric Indicators

The method `evaluate_semantic_response` evaluates the scenario and the response and produces six indicators, all of which have scores between 0 and 1 (Table II). If live mode is selected, then the Gemma model is queried via Colab to provide the numerical indicators. If fallback mode is used, then a lexical method searches for planning/sequencing terms, references to teammates or delegation, a sense of urgency, self-blame, catastrophization, and uncertainty. This approach is helpful because each indicator's name is based on a psychological term and not an internal feature.

TABLE II
PSYCHOMETRIC INDICATORS

Indicator	Source	Typical textual signals
Perceived control	PSS [3]	Concrete first action, ownership
Perceived threat	STAI [4]	Catastrophic or fearful wording
Emotional arousal	STAI / SRSI	Intense emotion words, urgency
Cognitive load	PSS [3]	Overwhelm, competing demands
Attentional flexibility	SRSI	Re-planning, adapting
Collaborative orientation	SRSI	Delegation, asking for help

D. Yeo-Network Mapping

The function `predict_7_attributes` uses the six indices to calculate seven attributes, one for each of the Yeo networks [5] because of their common correlation with the psychological function (see Table III). For example, high perceived control and high attentional flexibility will increase the Frontoparietal and Dorsal Attention attributes, while high arousal and threat will increase the Limbic and Somatomotor attributes. This mapping is made by design, based on existing literature [6], [7].

TABLE III
MAPPING OF INDICATORS TO YEO NETWORKS

Network	Function (summary)	Main indicator inputs
Visual	Visual processing	Scenario text (no direct indicator)
Somatomotor	Bodily arousal	Emotional arousal, threat
Dorsal Attention	Goal-directed attention	Attentional flexibility, control
Ventral Attention	Reorienting	Attentional flexibility, cognitive load
Limbic	Emotion and threat	Threat, emotional arousal
Frontoparietal	Executive control	Control, collaboration
Default Mode	Rumination	Cognitive load, threat

E. MLP Scoring and Benchmark Comparison

`run_mlp_model` function feeds seven attributes to the MLP regressor that consists of one hidden layer of eight ReLU units, using the Adam optimizer, mean squared error loss function and learning rate equal to 0.001 based on values from `tribev2.txt`, `tribev2_output.csv` and random data. It returns resilience score in the range 0-1. If the composite attribute is represented as a

ratio of control and arousal-related networks, it can be written as:

$$\text{Score} = (\text{FPN} + \text{DAN}) / (\text{FPN} + \text{DAN} + \text{SMN} + \text{LIM} + \text{DMN}) \quad (1)$$

where FPN, DAN, SMN, LIM and DMN stand for Frontoparietal, Dorsal Attention, Somatomotor, Limbic and Default Mode network attributes correspondingly. The targets of training determine what the score represents: the model is trained to replicate them, therefore its accuracy reflects the level of calibration mapping capture rather than ability to predict a person's actual behavior. After that, `compare_with_csv` function determines the differences between the score of candidate and benchmark scenarios according to seven criteria (stress handling, burnout resilience, emotional regulation, cognitive load, inclusivity, adaptability and collaboration). Then, `generate_behavioural_interpretation` function produces descriptions of differences regarding stress handling, burnout resilience, anxiety and emotion changes, and inclusivity and adaptability criteria.

F. Fallback, Error Handling and Human Oversight

Any problem with the hosted invocation such as timeout, expired tunnel, and error code is logged in the terminal on the server side but is not exposed to the user, with the pipeline running in local mode; and the output is marked accordingly for its mode. The submit endpoint is protected by an outermost exception handler in order that the server gives a proper response instead of an error. The API keeps the view of candidates separate from the evaluator interface, where each step, indicator, network value, score, and interpretation can be viewed. There is no automatic acceptance/rejection provided by the system.

IV. IMPLEMENTATION AND RESULTS

The prototype is run using an HP laptop having an AMD Ryzen chip, 8 GB of RAM, and 512 GB SSD, along with the hosted service on the Colab free-tier CPU runtime connected through a Wi-Fi link. The interface comprises of one dark-colored page (see Fig. 2) including a scenario navigation panel, a response box with a character counter, and a button for example responses. In case of submission, the live console first displays the names of all eight pipeline stages and then shows

the score badge, seven network bars, and interpretation cards (Figs. 3 and 4).

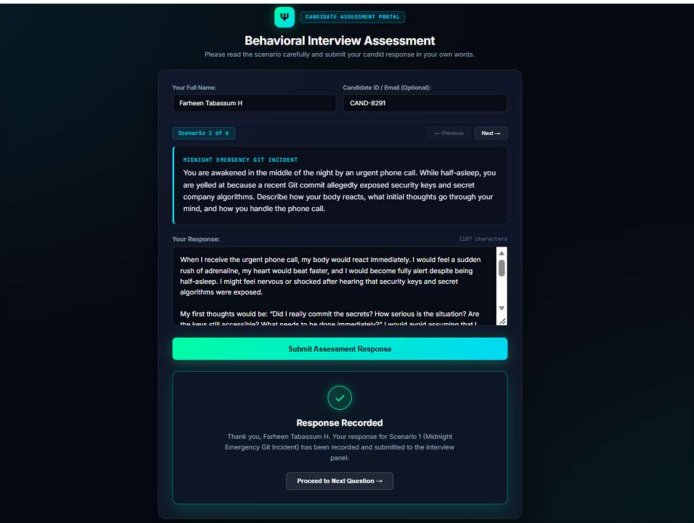


Fig. 2. Candidate examination interface showing the scenario and response box.

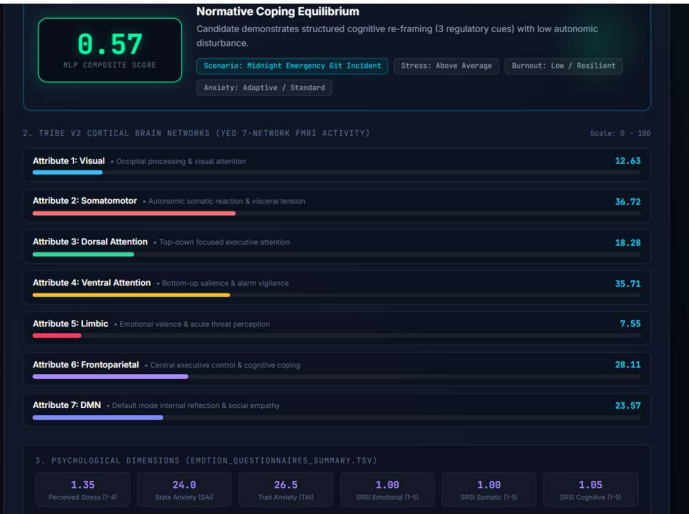


Fig. 3. Live console output: composite score of 0.57 and the seven Yeo-network attribute bars.

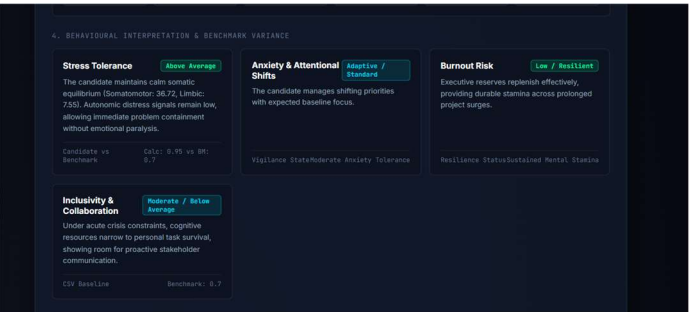


Fig. 4. Behavioural interpretation cards for stress tolerance, anxiety and attentional shifts, burnout risk, and inclusivity.

A. End-to-End Demonstration

Two answers of divergent style have been provided. Answer A (Acute panic and paralysis) revolves around the fear of consequences, responsibility, and the feeling of being in control; Answer B (Normative workplace response) outlines the initial step and distributes the tasks among the team members while acknowledging the uncertainties. Answer B scored higher at 0.57 as well as got rated for steady stress management with good adaptability while the answer A received the lower score at 0.28 and the profile was characterized by Somatomotor, Limbic and Default Mode network involvement and the presence of the cards representing the increased levels of anxiety and vulnerability to burnout (Fig. 5). Fig. 6 shows the comparison between one of the candidates and the benchmark scenarios. These two samples provide the reaction of the pipeline to different writings but not the statistical validation.

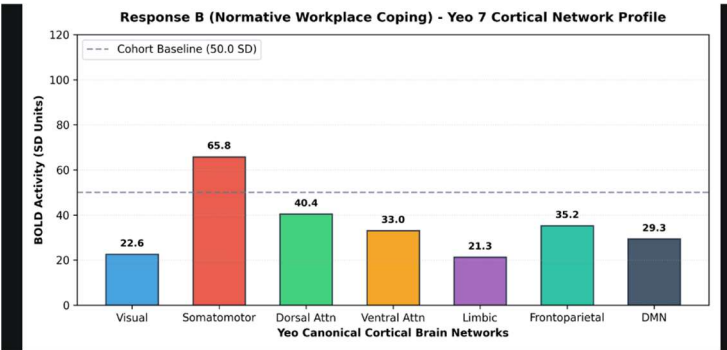
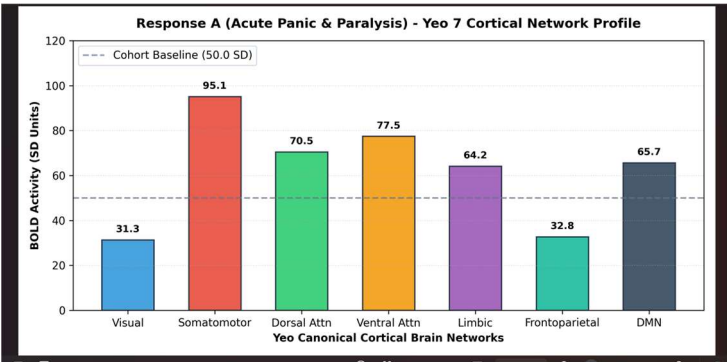


Fig. 5. Estimated Yeo-network profiles: (top) Response A, acute panic and paralysis; (bottom) Response B, normative workplace coping.

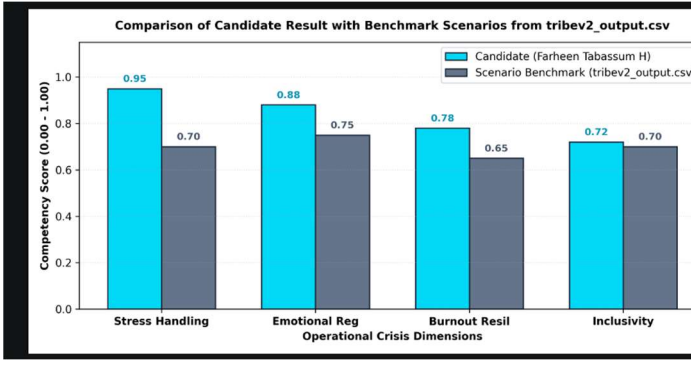


Fig. 6. Candidate result compared with benchmark scenarios from tribev2_output.csv.

B. Functional Validation

Reliability behaviors have been performed manually (Table IV). The most crucial test would be when the hosting service disconnects, and the user experience is not compromised, as the pipeline fallback ensures a result is still obtained along with the mode. All seven tests conducted passed the expectations.

TABLE IV
PIPELINE BEHAVIOR UNDER TESTING ENVIRONMENT

Test condition	Expected behavior	Result
Reachable Colab service	Live mode result with Gemma Success	Success
Invalid Colab URL	Fallback to local mode with valid result	Success
Timeout on Colab	Ten-second timeout calling and fallback	Success
Empty answer	Validation message, no crash	Success
long answer	Result delivered, no crash	Success
Unusual internal error	Error handled with JSON response	Success
Missing scenario files	Terminal error with safe default interface setting	Success

C. Quantitative Evaluation

MLP was assessed using six held-out samples isolated from the training set by using k-fold cross-validation. Coefficient of determination for prediction errors between the predicted values $\hat{y}_i$ and target values y_i is:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (2)$$

See Table V for the results. The network achieved a coefficient of determination $R^2 = 0.976$ across all samples with individual values ranging from 0.914 to 0.999 and a mean absolute error of 2.35 BOLD SDs. The median time delay from prediction input to output value was 1.18 seconds in live Colab mode and 0.02 seconds in local fallback mode, so the fallback method is also faster. Figure 7 illustrates training and validation loss plots. Since the target variable is an artificial calibration construct, a high R^2 indicates that the network is capable of predicting the

calibration mapping but does not imply prediction of any workplace behavior.

TABLE V
MLP EVALUATION RESULTS

Metric	Value
Held-out samples (N)	6
R^2 (overall)	0.976
Mean absolute error	2.35 BOLD SD
Per-network R^2 (lowest / highest)	0.914 / 0.999
Median latency, live (Colab) mode	1.18 s
Median latency, local fallback mode	0.02 s

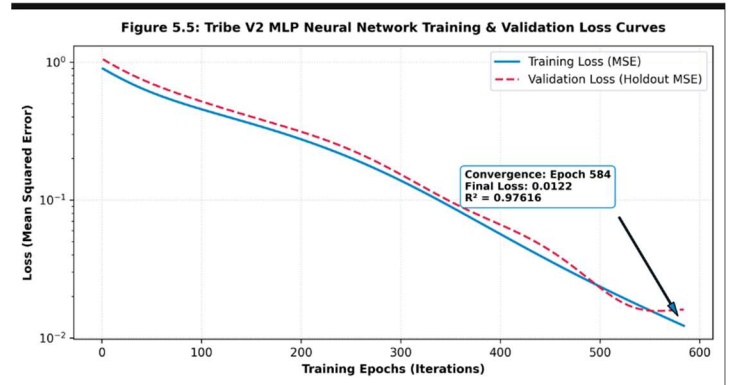


Fig. 7. MLP training and validation loss curves.

D. Discussion and Limitations

The findings provide preliminary evidence for the validity of RQ1 to RQ4 on the system level as the pipeline works from end to end, all transformations become evident, the failover makes the service always available, and the profiling and interpretation of the output are sensible in case of different responses. The process of development revealed a number of implementation problems, including the server's failure due to the absence of the llm module, expiration of free tunnel URLs and Colab sessions, possible leaking of raw stack traces to the user, the tunnel interstitial page making the API call impossible, and using an old request schema by the standalone client. All that resulted in creating llm.py, treating the hosted service optionally with fall back, safe JSON error messages, bypass headers with timeout, and the client's schema upgrade.

However, several caveats have to be noted clearly. First, there is no neuroimaging data concerning the examined candidates, that is, network measures are estimated interpretation of the data and should not be treated as either measurement or prediction of brain activities. Second, the accuracy of estimation in relation to the calibration data does not indicate predictive validity – there should be an independent

study of the scores correlation with certain outcomes (e.g., supervisor evaluation). Third, language-based scoring is subject to influence of writing fluency and language background and culture and, therefore, requires testing for fairness. Fourth, the size of the benchmark data set is quite limited, and the evaluation of six samples is indicative rather than definitive. Finally, the results of the fallback mode are obtained by a simplified heuristic analyzer and are not comparable to those of the live mode.

V. CONCLUSION AND FUTURE WORK

This paper demonstrated the Psychological Evaluation System prototype that transforms the candidate's written answer to the work-related crisis situation into six psychometric parameters, seven Yeo-networks' estimates and a resilience score with benchmarking and interpretation in simple terms. It works via the Gemma service hosted on Google Colab when available and a deterministic pipeline locally otherwise, providing thus a proper output always. MLP replicated its calibration data with $R^2 = 0.976$, and the prototype demonstrated how the profile and its interpretations depend on the contrasting inputs consistently. The primary value of the approach suggested is in making the rationale of the stress tolerance score explicit and open to discussion; it does not replace human evaluation or diagnostics.

Future research has four strands: validation of the criteria by measuring the same construct using independent criteria for stress resilience and job performance together with confidence intervals; fairness audit covering different writing styles, different languages and demographic subgroups; improved linguistic analysis involving calibration of the indicators with PSS and STAI scores and comparison of other embedding and linguistic models; and an ethical pilot study testing the relation between the network vocabulary and any physiological or imaging variable.

REFERENCES

- [1] F. P. Levashina, C. J. Hartwell, F. P. Morgeson, and M. A. Campion, "The structured employment interview: Narrative and quantitative review of the research literature," *Pers. Psychol.*, vol. 67, no. 1, pp. 241–293, 2014.
- [2] F. L. Schmidt and J. E. Hunter, "The validity and utility of selection methods in personnel psychology: Practical and theoretical implications of 85 years of research findings," *Psychol. Bull.*, vol. 124, no. 2, pp. 262–274, 1998.
- [3] S. Cohen, T. Kamarck, and R. Mermelstein, "A global measure of perceived stress," *J. Health Soc. Behav.*, vol. 24, no. 4, pp. 385–396, 1983.
- [4] C. D. Spielberger, *Manual for the State-Trait Anxiety Inventory (Form Y)*. Palo Alto, CA, USA: Consulting Psychologists Press, 1983.
- [5] B. T. T. Yeo et al., "The organization of the human cerebral cortex estimated by intrinsic functional connectivity," *J. Neurophysiol.*, vol. 106, no. 3, pp. 1125–1165, 2011.
- [6] V. Menon, "Large-scale brain networks and psychopathology: A unifying triple network model," *Trends Cogn. Sci.*, vol. 15, no. 10, pp. 483–506, 2011.
- [7] A. F. T. Arnsten, "Stress signalling pathways that impair prefrontal cortex structure and function," *Nat. Rev. Neurosci.*, vol. 10, no. 6, pp. 410–422, 2009.

- [8] M. E. Raichle et al., "A default mode of brain function," *Proc. Natl. Acad. Sci. USA*, vol. 98, no. 2, pp. 676–682, 2001.
- [9] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP-IJCNLP*, 2019, pp. 3982–3992.
- [10] S. d'Ascoli, J. Rapin, Y. Benchetrit, T. Brooks, K. Begany, J. Raugel, H. Banville, and J.-R. King, "A foundation model of vision, audition, and language for in-silico neuroscience," *arXiv preprint arXiv:2605.04326*, May 2026.
- [11] S. A. Adedokun, D. A. Adedokun, B. O. Ishola, R. I. Adeniran, and C. O. Olaleye, "Agentic AI and autonomous decision-making: A review of human-in-the-loop frameworks, oversight mechanisms, and trust calibration," *Int. J. Res. Innov. Appl. Sci. (IJRIAS)*, vol. 11, no. 3, pp. 1350–1374, Mar. 2026.
- [12] E. C. Cheng and J. Cheng, "A decoupled human-in-the-loop system for controlled autonomy in agentic workflows," *arXiv preprint arXiv:2604.23049*, Apr. 2026.